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Building Consistent Formal Specification for the Service Enterprise Agility Foundation

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ABSTRACT

Contemporary service enterprises face challenges of the rapidly evolving technologies and business environments. Thus, the precise analysis and the accurate planning can be a key to the selection of the "right" strategies for IT investments and the implementation of any new technologies. Consequently, the purpose of this paper is to contribute to the profiling and development a "Fail-safe" (i.e., Consistent) Service-oriented formal model to support architectural decisions in service enterprise planning, and dynamic business activities managing by "Fault-tolerant" design. Furthermore, the objective is to construct a "process/formal model" that can be applied by services providers in assignments concerning enterprise architecture, or IT architecture, planning and development. And, in turn, the goal of the "process/formal model" is to cover both areas of activity: the planning and enterprise processes effective management and development with a dependable (i.e., "Fault-tolerant") methodology to save time and resources along the life-cycle.

KEYWORDS

Consistent Process/Formal Model, Dependable Planning and Managing Methodology, Enterprise Agility Foundation, Service Enterprise.

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1. INTRODUCTION

It is argued in (Ross et al. 2006) that individuals can carry out complicated responsibilities “almost automatically”, without thinking about their extreme complexity. Also, Ross et al. 2006 state that with increased experience and knowledge humans are able to achieve more and more complex goals. On the begin however, certain attention and efforts are still needed, nevertheless they turn out fast to be a “second nature.” And, after a while some individuals can succeed with exceptional skills. This becomes possible by humans because of routine repeating of tasks. Then, these routine tasks’ repeating transforms into the expertise and can be performed “almost automatically”, i.e. without additional thinking. As further step Ross et al. point out-“Because experts need not focus on routine activities in their field, they can concentrate on achieving greatness.” This theme is considered and deliberated also in (Pulkkinen 2006, Armour and Kaisler 2001, Andrews 1994).

Consequently, let’s compare this to the business activities performed at the enterprises using Ross et al. ideas. They claim that enterprises cannot compete with human brain intelligence. “Activities as simple as sending an invoice, taking an order can easily go wrong -even after considerable practice. To focus management attention on higher-order process, such as serving customers, responding to new business opportunities, and developing new products, managers need to limit the time they spend on what should be routine activities. They need to automate routine tasks, so those tasks are performed reliably and predictably without requiring any thought” (Ross et al. 2006). The (Pulkkinen 2006, Armour and Kaisler 2001, Prahalad and Hamel 1990) have the same belief about the development, while broadening the discussion by different aspects.

As a result of the analysis, Ross et al. 2006 introduces a new expression term “*foundation for execution*.” According to these researchers concept “the foundation for execution digitizes … routine processes to provide reliability and predictability in processes that must go right.” Moreover, they claim “the best companies go beyond routine processes and digitalize capabilities that distinguish them from their competitors.”

A comparable concept is argued in parallel by (Pulkkinen 2006, Armour and Kaisler 2001, Andrews 1994, Prahalad and Hamel 1990). These above authors come to the same viewing platform in the service enterprises development.

If to follow Ross et al. 2006 further "⋯ a '*foundation for execution*' is the IT infrastructure and digitized business processes, automating a company's core capabilities. ⋯ Building a foundation does not focus only on competitively distinctive capabilities, it also requires rationalizing and digitizing the routine, every processes that a company has to get right to stay in business."

A plethora of research is done onto the studying of the "*foundation for execution*" (Pulkkinen 2006, Armour and Kaisler 2001, Andrews 1994, Prahalad and Hamel 1990, AbouRizk et al. 1994, Bacon 1992, Barua et al. 1995, Whitman et al. 2001). As a consequence, there comes a question: Why is the "*foundation for execution*" so important for the prosperous service business?-Again, according to Ross et al. 2006 as well as (Pulkkinen 2006, Armour and Kaisler 2001, Whitman et al. 2001) with the Internet appearance and dramatic increase by it of the opportunities for the inter-enterprises collaboration, it was uncovered in the same time an rigidity and an incapability to adapt to new channels, processes and technologies. This incapability was a consequence of complexity of proprietary closed enterprise IT legacy systems. "⋯ The complexity has not added value. Implementing standardized, digitized processes carries costs; particularly those associated with organizational change, but the benefits are simpler technology environments, lower-cost operations, and enhanced agility. ⋯" (Ross et al. 2006).

As a result, the concluding effect of an extensive research on "*foundation for execution*" done in (Ross et al. 2006) as well as (Pulkkinen 2006, Armour and Kaisler 2001, Whitman et al. 2001) is that despite we cannot foresee easily what will come or be changed, we still can figure out what remains unchanged for a longer period. For the enterprises it means that they can automate not-changing processes and activities, and in the same time concentrate on the future plans, turning in this way *"foundation for execution"* into *<foundation for agility>*."

2. BACKGROUND AND CURRENT SITUATION

2.1 Enterprise Architecture and Business Agility

By the definition (TOGAF 2002), the "enterprise" is any collection of organizations that has a common set of goals and/or a single bottom line. In that sense, an enterprise can be a government agency, a whole corporation, a division of a corporation, a single department, or a chain of geographically distant organizations linked together by common ownership.

The term "enterprise" in the context of "enterprise architecture" can be used to denote both an entire enterprise, encompassing all of its information systems, and a specific domain within the enterprise. In both cases, the architecture crosses multiple systems, and multiple functional groups within the enterprise (Armour and Kaisler 2001, Wegmann 2003).

The uncertainty arises from the evolving nature of the term "enterprise." An extended enterprise nowadays frequently includes partners, suppliers, and customers. If the goal is to integrate an extended enterprise, then the enterprise comprises the partners, suppliers, and customers, as well as internal business units.

Large corporations and government agencies may comprise multiple enterprises, and hence there may be separate enterprise architecture schemes. However, there is often much in common about the information systems in each enterprise, and there is usually great potential for gain in the use of a common architecture framework. For example, a common framework can provide a basis for the development of an architecture repository for the integration and re-use of models, designs, and baseline data.

The definition of an architecture used in ANSI/IEEE Std 1471-2000 is: "The fundamental organization of a system, embodied in its components, their relationships to each other and the environment, and the principles governing its design and evolution." The architecture can have two meanings depending upon its contextual usage:

1. A formal description of a system, or a detailed plan of the system at component level to guide its implementation;
2. The structure of components, their inter-relationships, and the principles and guidelines governing their design and evolution over time.

An architecture description is a formal description of an information system, organized in a way that supports reasoning about the structural properties of the system. It defines the components or building blocks that make up the overall information system, and provides a plan from which products can be procured, and systems developed, that will work together to implement the overall system. It thus enables to manage the overall IT investments in a way that meets the needs of the business.

An architecture framework is a tool that can be used for developing a broad range of different architectures. It describes a method for designing an information system in terms of a set of building blocks, and for showing how the building blocks fit together. It contains a set of tools and provides a common vocabulary. It should also include a list of recommended standards and compliant products that can be used to implement the building blocks.

An architecture design is a technically complex process, and the design of heterogeneous, multivendor architectures is particularly complex. Those, IT customers who do not invest in enterprise architecture typically find themselves pushed inexorably to single-supplier solutions in order to ensure an integrated solution. At that point, no matter how apparently “open” any single supplier’s products may be in terms of adherence to standards, the customer will be unable to realize the potential benefits of truly heterogeneous, multi-vendor open systems.

Typically, architecture is built-up because key managers (i.e., enterprise’ board of directors, CEOs, CIOs) have concerns that need to be addressed by the IT systems within the organization. Such people are commonly referred to as the “stakeholders” in the system. The role of the architect is to address these concerns, by identifying and refining the requirements that the stakeholders have, developing views of the architecture that show how the concerns and the requirements are going to be addressed, and by showing the trade-offs that are going to be made in reconciling the potentially conflicting concerns of different stakeholders. Without the architecture, it is highly unlikely that all the concerns and requirements will be considered and met (TOGAF 2007, King 1995, Schekkerman 2003).

The role of Enterprise Architecture (EA) has been evolving. When it was conceived in the 1980s (marked by John Zachman’s path-breaking work), EA had an IT-centric function. The focus was on organizing frameworks and technology products. The enterprise-wide picture was documented rather than planned. IT had a crucial role as a platform for the efficient and effective operation of an organization’s business. As more companies adopted IT to improve performance, it became evident that systems were built in functional silos with limited visibility of whole picture. This restricted the flow of information and impacted its integrity, adversely affecting collaboration across the organization (Pernul et al. 1998). The monolithic nature of these systems constrained organizational change at the technology, information and process levels. Removing these obstacles through improved business-IT alignment became the task of the late 1990s, a task that continues to keep IT organizations busy (Pulkkinen 2006, Armour and Kaisler 2001, Aziz and Obitz 2007, Perks and Beveridge 2003, Obitz et al. 2007).

The crucial role of EA is to provide a strategic perspective beyond day-to-day tactical project pressures. But, the survey found that 36% of all EA teams spend less than 30% for these activities and are, therefore, at risk of losing an enterprise-wide strategic focus. Marketing and communications is crucial when influencing a large organization. The lack of a structured multi-channel communication strategy results in a serious awareness gap concerning the architecture vision. The lack of appropriate metrics also makes it difficult to manage the IT assets of the organization as a portfolio (Wegmann 2003, Obitz et al. 2007, Iansiti and Favaloro 2006).

Besides, the enterprises must respond to changing customer needs with flexible systems and processes. Thus, the most cited objective of Enterprise Architecture (EA) is flexibility of business and processes to enable the enterprise to stay competitive, e.g. be agile.

Agility is the objective, not just for IT, but also for the entire organization. An enterprise that possesses the capability of sensing and responding business situations in a timely fashion is coined as a real-time enterprise, or highly agile. Unfortunately, most enterprises are slow responding to both problems occurring in their organizations and the changes of requirements from customers. Static processes that cannot adapt to changing needs are a liability.

Furthermore, defined as the ability to achieve competitive advantage by reconfiguring resources in response to business opportunities and threats, business agility is a critical success factor for corporations aiming to win in the flat world. And, the architectural approaches can help create agile organizations. This discipline was developed while modularizing IT landscapes to recombine the capabilities of systems that support changing business models and processes. Restructuring of organizations can be achieved by: modularizing the business into units, understanding the services provided by these units, and analyzing inter-relationships amongst them. By incorporating these techniques, enterprises can make their operational platforms more efficient, while adapting to changing customers and partners and their expectations (Aziz and Obitz 2007, Saha 2007, Bernus et al. 2003).

However, a technology-centered view of the business agility is still incomplete. True agility can only be achieved if companies rethink the business models they use today. The technology remains an essential building block of an agile enterprise, but more is required (BearingPoint 2005).

The challenges associated with enhancing business agility may seem modest on the surface, but in fact they can be substantial in financial, operational, and even cultural implications. The practical implementation of a comprehensive approach supporting business agility can pose a number of challenges before, during, and after implementation:

- The risk of inaction: There is a real temptation to delay the acquisition of a new technology solution in anticipation of future improvements. Unfortunately, the attitude delays the delivery of potential benefits of new technology that makes new business practices possible. It is difficult to calculate the opportunity cost of inaction, but that doesn't mean that those costs don't exist. In a similar way, it is not always possible to quantify the financial benefits and improvement in operational excellence that come from a proactive approach made possible by a flexible information system.
- Natural resistance to change: This can be a challenge when it comes from customers and employees, but it can be especially difficult to overcome if senior management shows anything less than full support. The challenge is the greatest in the most successful companies. After all, past practices have proven effective, so why change them? The countermeasure is to point out how expanding opportunities will improve company prospects with greater rewards for everyone.
- Potential for lost opportunity and inevitable finger-pointing: There is a risk of lost business and potential disruption whenever a business is in transition with new business practices and information resources being put into place. Building redundancy into a new approach can help ease a transition, but the IT department will still be blamed for not implementing new solutions fast enough. Likewise, not every idea for infrastructure improvement on information delivery will be implemented, with disappointment the

natural result, especially as people realize that a new solution will not solve every problem.

- Danger of losing critical institutional resources: In an effort to streamline and move toward increased flexibility, there is a risk of discarding some critical connections of information delivery. It is important to eliminate the collection and delivery of unnecessary information, or information is provided too often, or with too much detail. Both management and users need to be directly involved in the editorial process (Boggs 2007, Davenport and Short 1990, Silver 2002, Nigam and Caswell 2003).

2.2 Business Performance Efficiency/Agility Challenges Exploration

Service Enterprises have to examine regularly the effectiveness of their business and IT operations to identify opportunities for greater efficiencies. And, therefore, a Business Performance Management (BPM) is emerging and a critical discipline to enable companies to become real-time enterprises. BPM applications can promote an adaptive strategy by emphasizing the ability to monitor and control both business processes and IT events. By coordinating the business and IT events within an integrated framework, decision makers can quickly and efficiently align IT and human resources based on the current business climate and overall market conditions. Business executives can leverage the results of core business process execution to speed business transformation, and IT executives can leverage business views of the IT infrastructure to recommend IT-specific actions that can drive competitive advantage.

However, most BPM processes and architectures are usually linear and rigid; and once done, will be very hard to change. To change the requirements of BPM is sometimes like building a completely new application, which costs time and money. One of the main complexity stems from the fact that business level of concerns and information technology (IT) level of concerns are intertwined. Thus, it is often difficult to identify a clear delineation of where the business concerns end and IT concerns begin using the traditional software development methodology (Jeng 2006, Haeckel 1999, List et al. 2002).

Indeed, at the heart of every business is a complicated net of processes that form the foundation for all operations. These business processes are the lifeblood of the organization and typically include all of the humans and systems that exist within the enterprise. Since they play such a central role, business processes must be as efficient as possible to make the business as effective as possible. As a result, finding ways to automate and improve business processes has become a major focus for today's organizations as they struggle to find ways to become more agile and responsive to changing business climates.

In fact, an entire market of business process management has grown out of the desire to improve existing business processes and build new processes and services that will different-tiate a business from its competitors. BPM solutions aim to provide enterprises with a common platform that can tap into all resources, both human and system-based, to create, manage and optimize effective business processes that span the enterprise. BPM solutions can help organizations to maximize their existing technology and human infrastructure by linking existing systems and automating tasks that can free humans up to add value elsewhere within the enterprise (Ashton and Kelly 2006, Sinur et al. 2003).

Given its potential positive impact, it's hard for BPM not to sound attractive to most companies. But how to take the next step and begin to build a business case for making an investment (in time, resources and money) in a BPM solution is a noteworthy challenge.

One of the initial steps to making the business case for a BPM project is to categorize the potential business problems and actual costs that an organization faces if they continue with their current business practices. Thus, we identify and categorize in this project a number of problems and suggest the impact they can have on the business. In many organizations, both the immediate problems and potential risks are difficult to assign accurate financial figures to, although most managers find they can typically define a reasonable level of value or importance to each factor. For this step, we identify first which of the problems are the most critical to the business and then define their potential impact on the business (Table 1) (Ashton and Kelly 2006, BPMI 2012).

Once all of the potential problems and their associated costs have been identified, categorized, and quantified whenever possible, the next step is to identify and understand the various components that comprise an investment in business performance agility manage-ment (Ashton and Kelly 2006, Sinur et al. 2003, BPMI 2012).

Table 1. The Most Critical Business Performance Management Challenges and Their Potential Impact on the Business (Ashton and Kelly 2006).

Problem	Impact on business and potential costs
Inefficient processes	Potential inefficient process costs: - not completing processes on time; - loss of employee productivity/time for value-add work; - lost revenue opportunities/opportunity costs; - waste of precious human resources.
Slow response to business change	Potential response costs: - loss of existing business; - loss of new business opportunities; - failure to keep pace with competitors.
Expensive business errors	Potential business error costs: - loss of existing business; - costs to fix errors; - negative impact on reputation.
Lack of insight	Potential lack of insight costs: - opportunity costs; - inefficient processes - inability to optimize processes; - increased compliance risks; - lack of control.
Compliance	Potential compliance costs: - penalties from management and government or regulatory agencies; - loss of certification; - damage to reputation; loss of economic impact.
No services-based development	Potential costs include: - increased cost of development; - lack of responsiveness, agility; - loss of business opportunities.

2.3 Business Performance Management and Service-Orientation Complications

Today's service businesses depend on electronic processes at every level. An organization's ability to stay competitive relies heavily on being able to adapt its electronic processes in support of initiatives designed to improve productivity, reduce costs, deliver higher-quality information, and accelerate routine tasks. However, adapting these business-critical electronic processes requires evolving the systems they run on quickly and cost-effectively. This, in turn, is an extremely multifaceted task.

Service Organizations adopt new technologies over time, and many enterprises own systems and software decades old, as well as the latest technologies. The old systems typically run some of the enterprise's most critical processes, that is why they cannot be replaced. The new technologies are purchased to provide leading-edge capabilities. The result is often an incredibly complex environment composed of incompatible and proprietary information technology assets that were never designed to aggregate data or enable collaboration (Demydov et al. 2009). These include:

- *Legacy applications*: a wide range of them, written in a variety of procedural languages that typically do not have well-defined interfaces for collaboration with other applications;
- *Object-oriented applications*: many different applications developed as components for other applications, such as Enterprise JavaBeans™, COM, or CORBA objects;
- *Transaction systems*: custom-developed applications that run under the control of transaction processing monitors such as customer information control system (CICS), IMS Transaction Manager (IMS/TM), and other software; and whose individual transactions are well suited to incorporation into collaborative business processes;
- *Packaged application systems*: proprietary systems with well-defined, often completely proprietary, collaborative interfaces;
- *Databases and files*: proprietary relational database management systems (RDBMSs), legacy database management systems (DBMSs), and file systems with varying degrees of standard and proprietary interfaces;
- *Communication transports and message formats*: industry-standard transports such as hypertext transport protocol (HTTP), file transfer protocol (FTP), and simple mail transfer protocol (SMTP); proprietary messaging and queuing systems; and a variety of formats for exchanging data between enterprises, such as electronic data interchange (EDI), Society for the Worldwide Interbank Financial Telecommunication (SWIFT), Health Insurance Portability and Accountability Act (HIPAA), and others.

Taking into consideration all above, we can state that the business performance effectiveness is completely dependent upon the underlying software systems and applications to provide the components or services needed to support rapid orchestration (Veryard 2007, McKendrick 2007, Davenport 1993, Reijers 2006).

Going into further details, until recently, the largest enterprises have custom-developed new solutions or engaged system integration experts to connect existing information assets for supporting new initiatives. However, these efforts are costly, time-consuming, and sometimes impossible. Today's service enterprises must be able to adapt quickly, using their existing information technologies and avoiding the cost of ripping, replacing, or rewriting every incompatible technology that impedes business agility. They must be able to build a foundation for new agile services by exposing quickly existing systems as fine-grained services without having to write extensive code. These services must also be available across various channels in multi-protocol environments in order to allow organizations to adapt existing applications and systems to accommodate new business processes.

Service-oriented architecture is a new approach that helps organizations weave together diverse information technologies in a way that assures each technology's independent functionality while simultaneously enabling them to collaborate effectively within large-scale business processes, without having to scrap the investment or custom-develop massive amounts of code.

SOA offers protocol independence, meaning that different consumers of computing services-such as an application, a server system, or a human end user can communicate with the same service in a different way to obtain the data or functionality desired.

Services themselves function as discrete components, designed to aggregate underlying complex computing interactions into reusable "packages" that can be invoked whenever that particular piece of functionality is required.

Still, commonly IT organizations prefer to standardize their systems and create enterprise reference architectures, resulting in protocol dependence. In practice, this approach is often difficult to implement and maintain, especially when the next "new standard" inevitably arrives (Senor 2005, Acharya et al. 2005, Brezany et al. 2003).

Further mistake done by business leaders, when they hear about a new technology solving a business problem, is the "set-in-stone" scenario. A new application takes months to develop, and then it is released to the business for use. Unfortunately, since the whiteboard discussions six months earlier, the business processes have often changed. Changing the application, however, may take another two months, leaving the business leader with an outdated application and possibly impacting the company's ROI. Therefore, in the drive for business agility, business performance management should be based on the services-oriented principles (Ashton and Kelly 2006).

However, while SOA has many advantages, at the same time, it also has a number of weaknesses. High-quality IT development techniques become even more important (Kryvinska et al. 2009). Here are some specific challenges that SOA can make even more significant:

- *Governance*: first-class SOA demands first-class governance. Who is responsible for developing services? Who prioritizes what to develop or improve next? Who pays for the development, especially when multiple departments benefit?
- *Complexity*: composing applications of independent and loosely coupled services increases complexity. The more parts there are to an application and the more machines they run on, the more things that can go wrong. When one part of an application fails, it's possible for the entire application to stop working.
- *Reuse*: What do we have and what does it do? The common functionality is often developed repeatedly by teams who are unaware of each other. They may not make their code reusable unless there is reason to do so (Ellmer et al. 1996).
- *Process*: SOA is a new way of thinking. SOA development is similar to but not the same as traditional object-oriented and procedural development.

These challenges should not be taken lightly. They have the potential to disrupt the most well intentioned projects (Woolf 2008, OASIS 2006).

3. RESEARCH REQUIREMENTS AND GOALS

Because of increasing competition and limited capital budgets, enterprises need to assess carefully every information technology (IT) opportunity to ensure that their resources are spent judiciously. Conventional wisdom holds that IT services have enormous potential. However, enterprises continue to question the benefits of IT services in conjunction with other corporate initiatives such as business process engineering, eBusiness, and enterprise resource planning. Despite the potential benefits derived from IT services investment, traditional capital budgeting models have failed to estimate true IT service values due to their inability to measure complex interactions between IT and organizational performance.

Thus, this paper intends to contribute to the incorporated IT services evaluation methodology that integrates strategic business and enterprise architectures planning, business processes engineering, and service-oriented computing with the support of IT service investtments (Lee 2004).

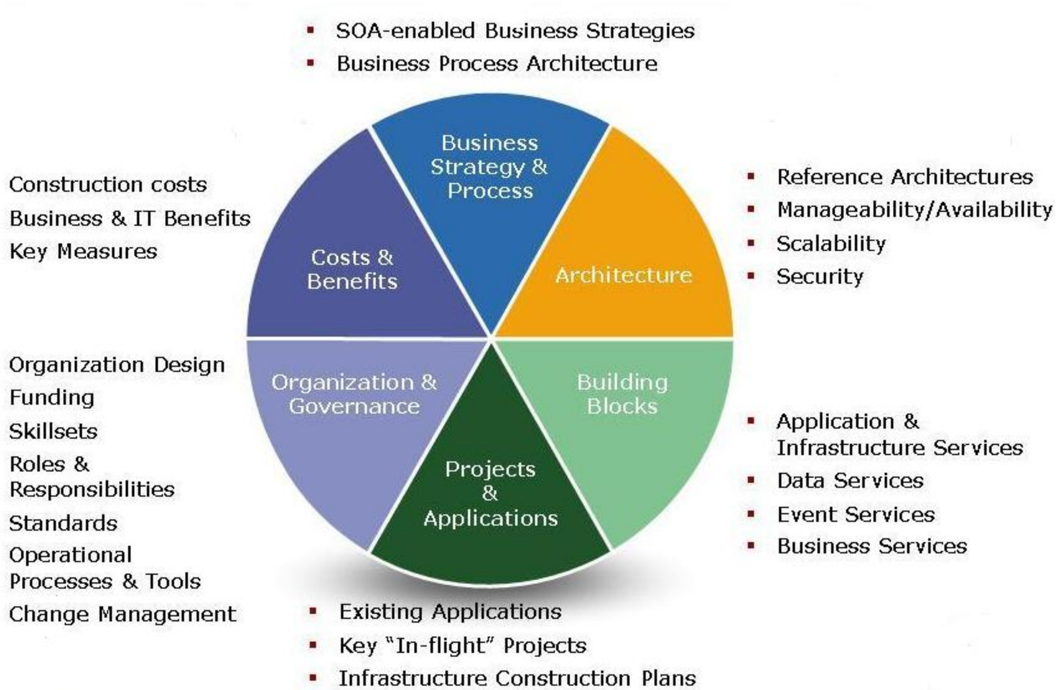

Figure 1. An Example of Domain-Based Architectural Planning Decisions Model

Following above, the main reason for the setting up and developing of enterprise agility foundation is the support of the enterprises by providing the fundamental technology and process structure for an IT service strategy. This, in turn, makes IT a responsive asset for a successful modern business service strategy. The effective management and exploitation of information through IT is the key to business success, and the indispensable means to achieve competitive advantage. Enterprise agility foundation addresses this need, by providing a strategic context for the evolution of the IT system in response to the constantly changing needs of the business environment. Furthermore, well-organized enterprise architecture enables to achieve the right balance between IT services efficiency and business innovation. It allows individual business units to innovate safely in their pursuit of competitive advantage. At the same time, it assures the needs of the organization for an integrated IT service strategy, permitting the closest possible synergy across the extended enterprise (Kryvinska 2010).

Consequently, the precise analysis and accurate planning is a key to the selection of right strategies and the implementation of any technologies. Thus, this paper contributes to a model for the running of architectural decisions in enterprise service architecture planning, and dynamic business service activities managing. First, decisions are made at the enterprise level, with strategic business considerations onto enterprise information, systems and technology strategy and service governance issues. The next step is to define the domains, to go on with domain service architecture decisions (Figure 1). And, at the systems level, the enterprise and domain architecture decisions are collected and converted into architecture descriptions accurate in precision, form and detail to be given as input to the information systems development process, following the architectural planning. The idea is to construct a process model that can be further applied both by IT end user service organizations managing their enterprise service architectures, and by services providers in assignments concerning enterprise architecture, or IT architecture, planning and development (Figure 2). And, the imperative goal is to cover both areas of activity: the enterprise service architecture planning and business performance management with a consistent methodology, to save time and resources along the process.

Also, important goal is an accurate determination of strategies and tactics for the successful implementation of Service-oriented technologies into enterprise architecture, in order to manage effectively running business processes and the development of new ones. Composition of software services is a fundamental part in sustaining enterprise business processes. Designed properly, executable processes can be used to support closely business performance by the integration of existing software services. In order to maintain business performance, the design of the executable process must closely follow the business events and activities, as perceived by business actors.

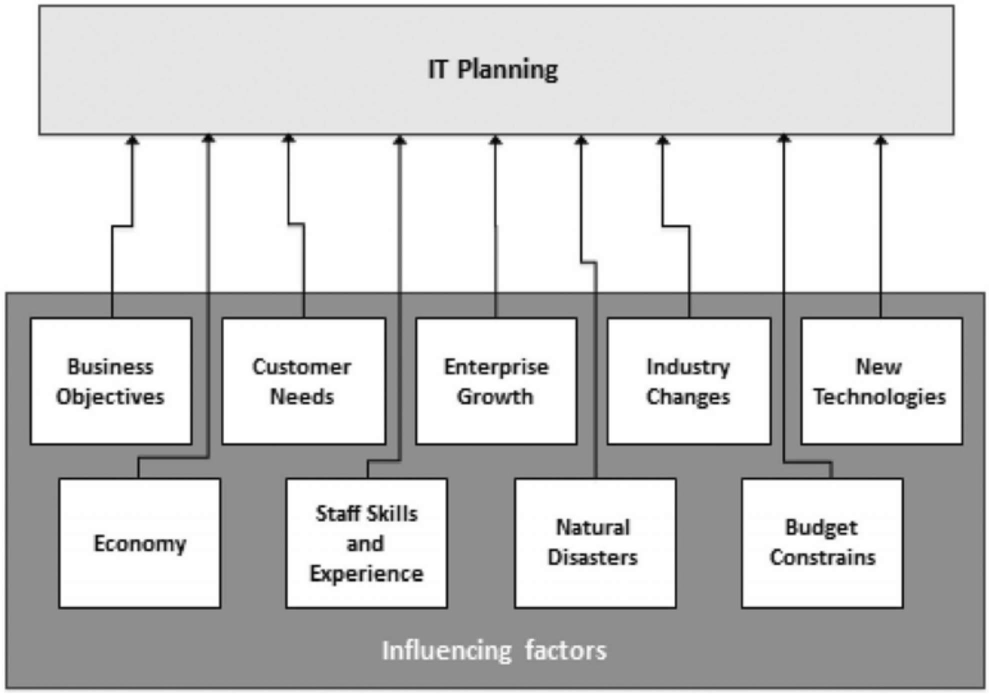

Figure 2. Influencing Factors by IT Planning

However, the design must also consider technical issues such as limitations in existing technology and systems. Thus, it is important to examine how technical system constraints influence the realization of business processes. Based on this examination, it is essential to present a set of realization types that describes the transformation from a business process into its realization. It is important also to examine the design criteria that need to be adhered to in order to cater to both business and technical essentials (Aziz and Obitz 2007, Iacob and Jonkers 2006).

As a next step, to understand the potential benefits of a service technologies investment, we believe that it is necessary to create a practical model to determine the value of the investment and the payback period. Traditionally, this has taken the form of a return on investment (ROI) model. However, there are many scenarios today where a traditional spreadsheet-based ROI model is not feasible or the specific quantitative values required are unknown. Demonstrating a positive ROI is the key factor in getting IT projects approved, and C-level executives are devoting significant attention to evaluating potential IT projects. But, the requirement to perform an ROI analysis is often perceived as a burden by IT divisions, which often lack the business analytical skill to perform effective analyses on their own, leading to the lack of value created by an expensive, time-consuming analysis process. The ensuring of the business/IT alignment is an overlooked value of employing ROI analysis. Putting ROI analysis in the context of business/IT alignment helps change it from a load to a powerful tool to help IT become more effective and more valued (Josefowicz 2003).

We suppose that, in order to develop a full-scale ROI analysis, it is necessary to create some sort of an alternative methodology that incorporates different aspects of ROI. It is also necessary to develop a category-based benefit analysis approach that can be used to build a compelling business case for IT/Services investments. And, important is to examine how an investment in business performance management will impact the organizations and their business operations. And, the model for building a business case can reflect the importance of balancing business priorities with an investment in IT/Services and offers organizations a way to begin weighing the costs vs. business impact (Ashton and Kelly 2006).

Furthermore, an important objective, concerning to the defined research subject, is the development of the model that can demonstrate how the service orientation can provide the business value.

Major goals of service orientation, in comparison with other software architectures used in the past, are to enable faster adaptation of software to changing business needs, and in turn-cost reduction in the integration of new services, as well as in the maintenance of existing services (Gartner 2007). So, borrowing from the investment theory of the "efficient frontier", which points to the maximum possible expected returns for given levels of risk, it is imperative to draft a model showing for any given technology, that there is a range of optimal choices in balancing the trade-offs between business value and IT costs. The service orientation is shifting the range of optimal choices and allowing for a new set of trade-offs and optimal decisions. The key point is, in the relation to pre- service orientation approaches, it moves the "efficient frontier" of IT investment out further (Melnicoff et al. 2007). However, without a structured approach to measuring business value organizations will struggle to secure the service-related projects funding necessary to drive further adoption.

On the other hand, most integration projects are authorized not to save IT costs, but rather to enable the development of demanding new businesses processes and services those must span multiple application systems. Enterprises are under pressure from a variety of external forces to implement integrated processes that finish faster, or provide more data than previous processes. However, no enterprise will upgrade its integration mechanisms (or make major service orientation infrastructure investments) just to clean up an undocumented, chaotic set of interfaces. Enterprises will only be motivated to consider new ways of integration when they have new applications (or a major change to existing applications) that will require significant spending on integration (Gartner 2007, Linthicum 2004).

As a result, the main idea behind this research is to leverage the above approaches to avoid setting unrealistic business value expectations, while also engaging clearly the business by extending IT-focused benefits into business focused metrics (Gartner 2007).

4. DEVELOPING A COHESIVE METHODOLOGY TO SUPPORT CONSISTENT SPECIFICATION/MODEL FOR SERVICE ENTERPRISE AGILITY FOUNDATION

In the relation to the defined research goals, this paper contributes to the cohesive model for the running of architectural decisions in service enterprise resources planning, and dynamic business performance managing, as well as analyzing of the IT services investment aspects for above mentioned activities.

The novelty of the work lies into the multidisciplinary approach by merging different research areas supporting the development of the service enterprise building blocks interoperability, as shown in Figure 3.

We have discovered during our initial examination that a lot of research has focused on the theoretical contributions to the service enterprise IT resources planning and management, but little attention has been paid to the evaluation methodology that integrates business performance and IT (Kryvinska et al. 2008b, Engelhardt-Nowitzki et al. 2011). Since IT services may fundamentally change the ways in which business organizations interact with internal and external constituents, business strategy and business performance have to be considered in the evaluation process. Besides, the methodologies for developing of service/ software architectures are still not obvious and not uniformly accepted (Engelhardt-Nowitzki et al. 2011, Auer et al. 2011). Corresponding to these constrains we are developing a methodological model that allows for analyzing and modeling of varied processes occurring onto different stages of the enterprise life-cycle.

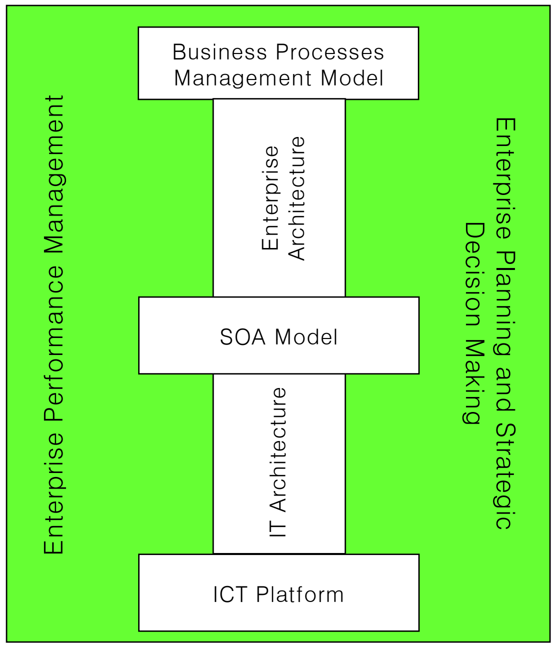

Figure 3. A Framework for the Service Enterprise Building Blocks Interoperability vs. "Foundation for Agility"

This methodology also assists to estimate the impact of the service enterprise resources planning and business processes design onto organizational performance, and helps to determine the most appropriate business processes and IT services to achieve business strategies. The evaluation process may iterate until the best IT and business process combination is developed.

We base our approach onto the framework known as a "foundation for execution" vs. "foundation for agility", which is the IT service infrastructure and digitized business processes, automating an enterprise's core capabilities.

We have already developed the beginning of the logic to address the difficulties within an incorporation of analyze and modeling of varied processes within the service enterprise (Kryvinska et al. 2008b, Kryvinska et al. 2011). Our goal in this paper is to fill the gap upon and between two points: extend existing theory and methods, and support them with useracceptable tools to be able to solve the obstacles observed in practice.

Thus, the proposed evaluation methodology (Table 2) goes through the main enterprise life-cycle phases:

(1) Strategic Analysis and Planning;

(2) Services Governance, Enterprise Performance Management;

(3) Migration Planning, Development, Managing Strategic Changes (Veasey 2001).

Table 2. A Framework for the Enterprise Life-cycle Management Strategies

Enterprise Life-cycle Phases	**Technological Aspects**	**Economic Aspects**
Strategic Analysis and Planning	Foundation for Agility	
Service Governance, Enterprise Performance Management		
Migration Planning, Development, Managing Strategic Changes		

To ensure addressing all important issues, an efficient methodology for the service enterprise agility foundation has to encompass of 8 Phases:

- **P1: Investigation and Comparative Analysis of Related Approaches:** perform an overall research in the area of service enterprise planning, business service models; examine related approaches and perform comparative analysis to form a predictive model of success-factors; investigate the potential application areas that may suggest interesting case studies.

- **P2: Requirements Gathering and Criteria Definition for the Effective Planning and Decision-making:** Every new planning decision has to be built onto the following requirements:
  a) clearing up of the mission (a concise statement of the existing or new project planning reasons),
  b) setting up the objectives (broad statements describing the targeted direction), and
  c) characterization of goals (quantifications of objectives for a designated period of time, in order of the portfolio planning).

And, the main criteria to identify if the decision made is efficient are as follows:

a) broad and transparent vision, and

b) agile strategies.

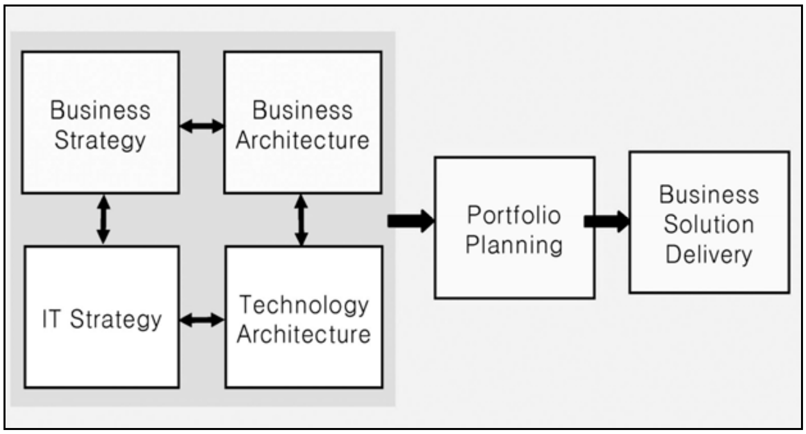

Figure 4. Planning and Decision-making Value Chain

Thus, the key output of this Phase is creating a specification (Figure 4) based onto the vision requirements (mission, objectives, goals) for further strategies (statements of how objectives will be achieved and the major methods to be used) and tactics (particular action steps that map out how each strategy will be implemented) development.

- **P3: Total Architecture of the "Foundation for Execution" Synthesis:** Before further detailed planning and realization, it is necessary to determine whether a project is feasible and to select an appropriate design without burning up all of the time and resources. Two interacting factors make this issue difficult to respond: the large number of possible designs and the fact that we cannot estimate costs and schedule until a design is at least partially completed. Because of large number of possible designs, it is generally hard to provide an exploration of all design alternatives, assess the ability of each to provide the expected benefits, and estimate costs to implement each design. What is needed is a way to simplify this search. Thus, it is important to use on this Phase the "Total service architecture synthesis" iterative method that applies "spiral development principles" (Brown 2007).

- **P4: Organizational Strategic Analysis and Planning in IT Services Investment:** It is important to know where IT services investment and decision-making methodologies fit into the general planning framework (Figure 2 and Figure 4). To understand their role, "hierarchical planning" approach will be applied here. There are three basic levels of planning in all organizations, and in all functional areas:
  1) strategic-,
  2) tactical-, and
  3) operational planning.

 And, with this approach, it is essential to go through the following steps:
  a) external analysis of competition and threats;
  b) internal analysis of the enterprise's strengths and weaknesses;
  c) overall corporate strategic planning; and
  d) functional area strategic planning (Lee 2004, Schniederjans et al. 2004).

- **P5: Service Enterprise Governance through the Enterprise Performance and Business Processes Management:** In order to build an agile framework for the enterprise governance (Figure 2), we use here a methodology based on the hierarchical approach of the enterprise management model. It is separated on three essential layers: an (1) enterprise performance layer (also called business performance management and corporate performance management), a (2) business process management layer, and a (3) SOA platform. We do not focus here onto the underlying ICT platform because of its business-support role.

 The enterprise performance management layer focuses on specifying the strategic key performance indicators of the enterprise and tying these to underlying business processes and policies, or business rules. Business processes, in turn, utilize the infrastructure services provided by the service-oriented architecture. And, the enterprise service bus (ESB) provides a common standards services-based brokering container. In its traditional role, the focus of IT is the underlying infrastructure and its reliability. New technologies such as ESBs have emerged to enhance, to extend, and to improve IT deployments. However, at their core the focus of these technologies is on low-level infrastructure. In contrast, business stakeholders focus on performance management, strategies, and key performance indicators of the business. At this layer IT infrastructure is viewed as an enabler. Business process management systems are the middle layer. Business process management systems allow enterprises to separate their business processes and business rules to model and to manage them independently of applications. This is the key.

Besides, BPM includes business process analysis and activity monitoring. Businesses are defined through KPIs, goals, objectives, policies, processes, practices, and priorities. The analysis component allows for modeling, simulating, and analyzing the performance of the processes. The analysis of historic data is the key in identifying potential bottlenecks in the processes or the participants (e.g., human, trading partner, or back-end applications) of the process activities, which in turn is the main task of this Phase (Khoshafian 2007).

- **P6: Consolidating Common Business and IT Functions through SOA:** Here, we come to developing of service-oriented enterprise models. It has been proven that SOA is an effective approach for building of large-scale applications. Thus it is crucial here to bridge the gaps between enterprise business performance modeling and SOA by providing a cohesive method of building service-oriented BPM systems. This method has to be refined via the following means:
  1) creating formal modeling framework so that validation and verification can be carried out during the modeling process;
  2) validating this work through empirical study and verifying the quality of the models and applications that are developed via this method;
  3) combining this approach with other development methodologies such as model-driven approach to leverage the best from different perspectives (Jeng 2006).

The modeling framework we build onto the *matrix method* that focuses on a business and IT focal points (Figure 5).

This Phase also has to cover one of the most important concepts in service orientation: namely, service oriented methodologies, including service oriented analysis and design. It contrasts traditional waterfall to iterative methodologies. Service oriented applications need to be developed in the context of iterative and continuous improvement methodologies. Furthermore, in the analysis and design of service orientation, there are some fundamental differences when compared to more conventional application development. These could be described through three dimensions. When we either build or interact with a service, that service will have properties along each of these dimensions:

1) leaf service or composition from other services: new services can be built from existing services;
2) exposing existing implementations or building new services: once services are built, they are consumed internally, or might be interests in publishing the interfaces of existing enterprise information systems;
3) services for internal or external requestors (Gartner 2007, Khoshafian 2007).

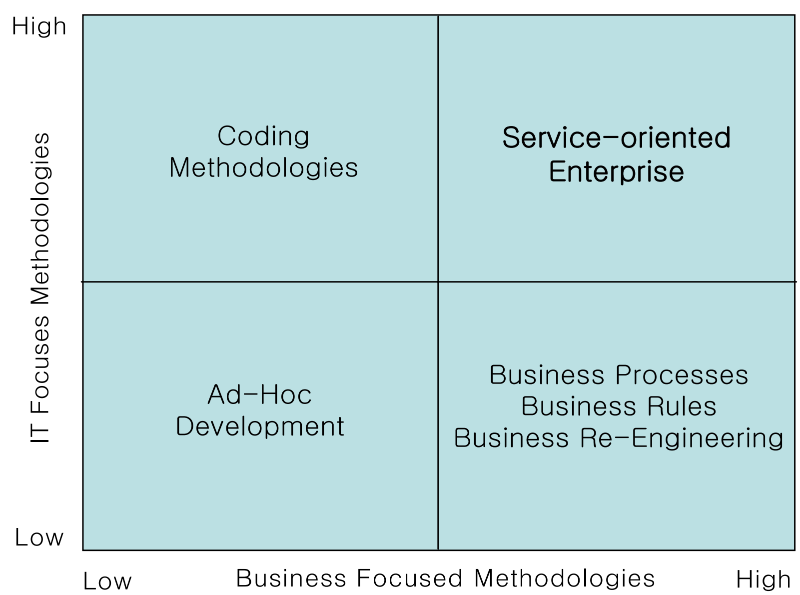

Figure 5. Matrix-based Modeling Framework

- **P7: Migration Planning, Development, Managing Strategic Changes:** This Phase engages a methodical analysis, development and determination of the inputs, outputs and business processes of the enterprise systems. This includes collecting cost and benefit information. The idea is to provide a base-line in which the measure the future impacts of change brought is influenced by changes we may make in IT now.

Also, this Phase involves the course of business process re-engineering, where we look at current policies, practices and procedures in delivering services to all customers (external customers and internal users), and see if a revision in those policies, practices, and procedures might lead to an improvement in service or productivity. It has to be performed here the configuration and functionality analysis that, in turn, depends on what is being considered in the change process. Usually it involves exploring of the alternative IT configurations. These configurations are examined in terms of how well they function to serve areas of business operations, such as marketing, sales, manufacturing, finance, accounting, maintenance, engineering, and human resources.

And, finally, this Phase is a vital step that summarizes the enterprise planning and re-engineering processes. While all the above steps indicate that feedback is possible to make revisions from the prior step, if necessary, this step is a final form to check against the goals and objectives set at all the strategic, tactical and operational planning stages. It is final check to make sure that the cost and benefits observed and expected in the new system benefits are achieved (Schniederjans et al. 2004, Ventana Research 2003).

It should be applied here a "Spiral method" that includes the original spiral model (Boehm 1988), and uses a cyclic approach to the development process. It captures the major spiral model features: cyclic concurrent engineering; risk driven determination of process and product; growing a system via risk-driven experimentation and elaboration; and lowering development cost by early elimination of nonviable alternatives and rework avoidance. It indicates that the spiral model is actually a risk-driven process model generator, in which different risk patterns can lead a project to use evolutionary prototyping, waterfall, incremental, or other subsets of the process elements in spiral model diagram (Boehm et al. 1997).

- **P8: KPI-, ROI- Criteria Analysis, and Business Value Assessment:** In this Phase, the analysis of a range of quantitative and qualitative IT investment methodologies must be applied. For example, multi-criteria methodologies like the "Analytic Hierarchy Process" (AHP) has be used to rank differing configurations with relation to their ability to provide enhanced customer service. As well as, "Meta-analysis" approach applied to assess the SOA business value.

Obviously, this Phase is a step in where major IT investment methodologies are brought to bare on selecting and evaluating the best alternatives. This includes architecture-wide choices; or/and very detailed individual component choices that make up the architecture (Schniederjans et al. 2004).

Namely, it is crucial to analyze and find:

1) the sequential-type of investment decision-making problems: complicating the decision process and multi-criteria compounding decision issues;
2) key performance indicators (KPIs) of Enterprise Performance Management: EPM is a market-space that is fraught with acronyms. Focusing on the acronyms themselves can be confusing. It is more important to understand what they represent and how a particular organization addresses the concept behind the acronym. A KPI is a metric that rises above other potential metrics as being truly indicative of the organization's performance. KPIs can be both financial and nonfinancial. A change in a KPI usually results in taking action to assist or improve business operations.
3) EPM methodologies employed by a specific enterprise: A methodology is a particular process, or method, by which the enterprise measures its performance. Methodologies come in many forms, and many organizations use multiple methodologies. Some organizations may use methodologies with a similar concept, but with a different name, as follows: Activity-Based Costing (ABC); Shareholder Value Add (SVA), also known as Economic Value Add (EVA); Balanced Scorecard (BSC); Six Sigma and Total Quality Management (TQM); other quality/productivity methodologies (Osterfelt 2002).
4) the impact of BPM initiatives: aiming to resolve business integration issues by providing an integrated framework, enabling an organization to understand the status of business processes across business and IT, placing that understanding in context against goals and trends and taking timely action to improve business operations (Haberer 2003).
5) key areas where enterprises can expect the ROIs on their business activities: to analyze here different ROI models to demonstrate the incremental ROI that BPM/SOA can provide; apply to the analysis an approach that combines benefits of tangible and intangible factors on understanding the overall ROI of business services architecture

(Greenbaum 2005).

6) business value assessment: to assess BPM/SOA business value via a multistage process: by first (1) understanding BPM/SOA capabilities and business priorities; then, by (2) selecting schemes that directly complement these capabilities and address business priorities; and, finally, (3) linking BPM/SOA enabled capabilities to value metrics by explicitly defining quantifiable business benefits. Understanding capabilities enabled by BPM/SOA is necessary for identifying and measuring value. The challenge is to translate intangible benefits (e.g., capabilities) into tangible benefits (e.g., metrics). This is a cyclical, or more precise-a spiral process (Leinbach 2000, Shank and Govindarajan 1992). The capabilities, typically defined in technology terms, can then be translated into relevant IT and business benefits. Benefits can then be translated into meaningful and measurable metrics. Next is matching these metrics to the key business opportunities, initiatives, threats, as defined by the business, and then the value measurement is feasible. Trying to make the leap directly from BPM/SOA principles to value measurement will fail (Gartner 2007).

To summarize, the use of the accurate, and efficient IT investment decision methodologies can support and assist on all phases in the overall enterprise life-cycle process: strategic analysis and planning, governance, enterprise performance management; and migration planning, development, managing strategic changes (Schniederjans et al. 2004). The key undertaking is to leverage different approaches to avoid setting of the unfeasible business value expectations, and at the same time also engaging clearly the business by extending IT-focused benefits into business focused metrics (Gartner 2007).

5. METHODOLOGY OUTCOMES

The key impacting results from the methodology are specified in the sequential (e.g., according to the phases defined) order as follows:

A. The roadmap for the systemization of the research efforts: generic, common framework for solution to build an adaptive enterprise;

B. The conceptual framework with associated formal and best practice techniques that can be used for the design, deployment, and management of the enterprises: specific frameworks to support the projects, use-cases, scenarios, test- and collaborative-experimentation (Kryvinska et al. 2008a);

C. Specification of architecture, strategies and techniques for the optimized implementtation:
   a. Document that describes a design, to demonstrate added value;
   b. Interoperability terms to assist enterprise designers and system developers in composing and adapting executable solutions;
   c. A service modeling and delivery methodology, and the integrated tools to support design operations and analysis of design results for the behavior models;
   d. case studies in which the above mentioned results are applied and evaluated;
   e. definition and development of the rules perception, how to act into the certain situation fast and effectively based onto Service Level Agreement (SLA) regulations.

D. The portfolio of planning tools and/or tool combinations supporting different formation activities along with the main features to be taken into consideration:
   - Capabilities model based on modern technologies and technical requirements;
   - Commercial availability;
   - Capabilities model to interrelate different planning tools among themselves and with the Operational Support Systems (OSSs);
   - Explicit documentation of models, inputs and results;
   - Commitment for periodical updates and maintenance;
   - Validation process for a range of cases.

E. Business performance management: the capability solutions blueprint for:
   - Unified data (event and metric) integration and business intelligence to assess the key performance indicators;
   - Event-driven performance management of business operations and IT infrastructures;
   - Integration modeling of business strategies, goals, processes, metrics, situations, decisions and actions;
   - Intelligent decision making and action management to render adaptive control upon business/IT operations;

- Conveyed business rules and analytics to improve business effectiveness, deploy business strategies, and enhance agility of business/IT operations;
- Advanced business optimization based on key performance indicators to respond to regulatory and emerging changes for reducing enterprise spending and improving operational excellence;
- Advanced forecasting and planning to improve the speed and quality of the response to business situations for improving the effectiveness of business and IT operations.

F. A major objective of this study-area is to evaluate how SOA can change the way new Services are created and managed, both from the technical and business perspectives, and see how loosely coupled integration approaches can reduce the complexity and hence the cost of integrating and managing distributed Service platforms. Thus, the outputs are:
  - Common model for Services;
  - Services identification, analysis, business modeling, and design methodology;
  - Services integration model;
  - Services governance organization and behavior, as well as reusability model;
  - Business case and return on investments model.

G. Business value (from the IT investments) assessment framework, to assist in strategic analysis and planning, governance, migration planning, development, managing strategic changes.

6. CONCLUSIONS

We tried to rationalize the motivation to our research by following the citation of Charles Handy:

> "*When change is discontinuous, the success stories of yesterday have little relevance to the problems of tomorrow; they might even be damaging. The world at every level has to be reinvented to some extent*."
>
> Source: Charles Handy, Beyond Certainty, Arrow Business Books, 1996 (Henbury 2000).

Accordingly, in the rapidly evolving business environments, the precise analysis and the accurate planning can be a key to the selection of the “right” strategies for IT investments and the implementation of any new technologies. Thus, we contributed in this paper to the profiling and development of the consistent Service-oriented formal model to support architectural decisions in service enterprise planning, and dynamic business activities managing. We also assembled a process model that can be applied by services providers in assignments concerning enterprise architecture, or IT architecture, planning and development. This model, in turn, covers the planning and enterprise processes effective management, and the development with a dependable methodology to save time and resources along the life-cycle.

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